

Building a unified science of Natural Behavior with Densely Overlapping Measurement Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

1. Mission

Develop and disseminate a new class of scientific instrument called a **Densely Overlapping Measurement Environment (DOME)** that records the complete perception-action loop of a behaving agent as a single calibrated, uncertainty-tagged measurement, and build the autonomous organization dedicated to the measurement itself and form the boundary object that will align perceptual and motor neuroscience, musculoskeletal biomechanics, mobile robotics, and embodied AI into a single convergent science of sensorimotor control.

1.1. A New Observable: The Sensorimotor Loop

Every agent, living or engineered, must solve the same problem - sense a thin slice of available energy in the environment through a limited set of imperfect transducers and on its basis push its body against the world to propel itself towards its goal. Information flows in, forces flow out; the brain exists to yank the bones around.

We can measure individual components of this loop with extraordinary precision progressing at the rate of our technology. Camera and IMU based motion capture can track the kinematics of the body, force plates measure the kinetic forces between the body and world. Outward facing cameras can measure the world, and inward facing cameras can track the eyes. For humans, EEG nets and surface EMG record course records of patterns of neural activity in the cortex and musculoskeletal system. In animal models, dense electrode arrays resolve the individual spikes of neurons and motor units.

And yet, despite the extraradainary precision and progress along these individual threads of measurement, access to the coherent whole representing the shared observational context that each of these tools illuminates remains elusive. Occasionally, a research team will reach accross disciplinary divides and stitch several threads together for heroic integrative studies, but the methods rarely escape the Methods sections of a small cluster of publications spanning roughly the timescale of a single PhD.

[TODO - fold next two paragraphs into 'unmet need' and 'vision' sections below] Each of these teams sought out to answer a question at the edge of their discipline and instead found themselves in the business of tool building. The build these tools because they know the answers to the questions that they seek lie in the output of the integrated tool that does not exist. In an institution that rewards the plating of seeds over the tending of gardens, the long and laborious work of maintaining a novel complex instrument quickly overwhelms the capacity of an academic lab. Eventually the nascent tool collapses under the weight of its own complexity, and what was meant to be a bridge between two domains of study instead fades into an evergrowing pile of academic abandonware.

We need a new organization that exists outside of the siloed domains of the Academia that dedicates itself to the MEASUREMENT, the development and dissertation of world-class, convivial tools and novel skillsets explore the landscape the capacity unlocks. That organization is the **FreeMoCap X-Lab [FMC-X]**. We have already changed the landscape human-focused research by bringing high-quality capture to over 15,000 researchers, clinicians, robotists, and animators across 152 countries. With the support of the NSF X-Labs program, we will extend our mission and change entire landscape of neuroscience, biomechanics and robotics.

1.2. A New Instrument: Building the DOME

We define a DOME as a densely instrumented region of real space engineered (at some aspirational extreme) to record every possible measurable channel of the agent–environment interaction at once. It is the opposite of a space telescope, an empty volume of of lenses focused inward at the mysterious place where the body meets the world.

In practice, individual DOMES will comprise a partial subset of available instrumentation, selected based on use-case. However, critically, **All DOMES produce the same output** regardless of the component instruments - a metrologically-grounded record of semantically-coherent canonical models **partially hydrated** by measurements from the available instruments [TODO - Define/invoke Sensor-Grounded Ontology thing here].

A motion capture recording produces the same reconstructed Head A mocap Skull is assumed to have Eyes, regardless of whether the subject was wearing an eye tracker to hydrate its Degrees of Freedom. Both a ferret and a human have a Skull wqchich maybe approximated as the same RigidBody defining segments of the body of a humanoid robot [Collaborator - SH, CH, JH, GN, AS] [TODO - *This robot stuff is redundant to below, I think. need consollidation*]. The reconstructed RetinalInput derived from reconstructions of the Body, Eye, and Environment of a human participant running across a rocky field [Ref-Collab-MH/KB] are precisely the same data models as those derived from a Marmoset leaping from branch to branch with Neuropixel electrodes emdedded in its SuperiorColliculus leaping to platform [Ref-Collab-AH/JY], a Ferret chasing a fictive prey with a Miniscope viewer mounted over of its PosteriorParietalCortex [Ref-Collab-BS], and a GuineaFowl running across a pneumatically actuated platform with EMG-electrodes implanted in its MedialGastrocnemius [Ref-Collab-MD].

[TODO - reference specific work building DOME-variants in flagship facility and fan-out of built-to-use variants to collaborator network. Mention targetting specific hardware based on practical and research-grounded needs - target Retinal input good enough to target ephys, needs build better eye tracker (dense sensor array for better eye data in general, and adding unmeasurable DoFs like torsion and accomdation for accurate retinal optic flow(CURL). Need world sensor to map environment, built into head-mounted world sensor array of eye tracker, later expanding to attach that scanner to drones. Auto-calibrating camera sensor array - break practicality plateaus of large capture volumes, allow for autonomously driven experiments on ideal camera configureations, and ai-driven autonomous collaborative experiments by intergrating drone-swarms in the scene (i.e. the drone swarms and camera arrays interact to test different capture configurations, training the drone-swarm control policies, all which preps for integrations into DOME-MOBILE later to ground inside-out sensors of IMU-suit and head-mounted world scanner)]]

1.3. The New Capability

Because every channel is spatially calibrated, temporally synchronized, uncertainty-traced within a unified sensor-grounded ontology, DOME records drop into modern reinforcement-learning and robotics stacks with minimal reshaping — giving embodied AI the real-world, uncertainty-bounded sensorimotor corpora that safe learning requires and that no current dataset provides.

Internally, each sensor stream is modelable agaist every other stream, both within recordings (modelling eye movements against head movent to model VOR) and across recordings (back-fill old datasets with models of Cyclotorsion after we build a new eye tracker that measures torsion). The data also provides internal automatic error signals - downstream estimates of body kinematics fused from IMU- and Camera-mocap can back project onto the the original

camera images, defining a reprojection-error signal that can improve image pose-detection algorithms that are currently stuck on an increasingly long-in-the-tooth [TODO - synonym] COCO dataset.

The same **Sensor-Grounded Ontology** that [...allows us to align outputs for heterogeneous DOMES...] makes the instrument legible to a human — explicit entities, explicit relationships, no assumed domain knowledge — makes its records natively legible to a machine: typed, grounded data over which a model can build reliable reasoning chains rather than guessing at unlabeled streams. This cleanliness is not a layer added on top; it is the property that lets a DOME route its live streams and logs through a local model to assist novices, experts, and developers alike, and that lets published results tag back into one shared entity-relationship structure connecting them to the wider body of science.

1.4. Unmet Needs

[TODO - this section is redundant and overlapping with other stuff, and also way too verbose. Can be cut down later] An instrument like this is unmet by existing organizational structures and funding mechanisms for reasons that are structural, not incidental. Building a scientific instrument is a **distinct craft, not a downstream application of domain expertise** — mastery of electrophysiology or biomechanics does not transfer to the engineering of a calibrated, maintainable measurement system, and because that difficulty is invisible from outside the craft, integrated measurement is repeatedly scoped as a student side-project, under-resourced, and abandoned when it proves to be a career of engineering rather than a semester of it. The deeper failure is not that these efforts collapse but that **even their successes do not compound**: a research result and a research instrument are different deliverables, and the second — a one-off tuned to a single rig, undocumented and unmaintained — rarely survives the trainee who built it. The field rewards the demonstration, not the durable artifact, and often cannot tell them apart, so proofs of concept are celebrated as finished tools and the unglamorous engineering that would turn one into an instrument others can trust is never done.

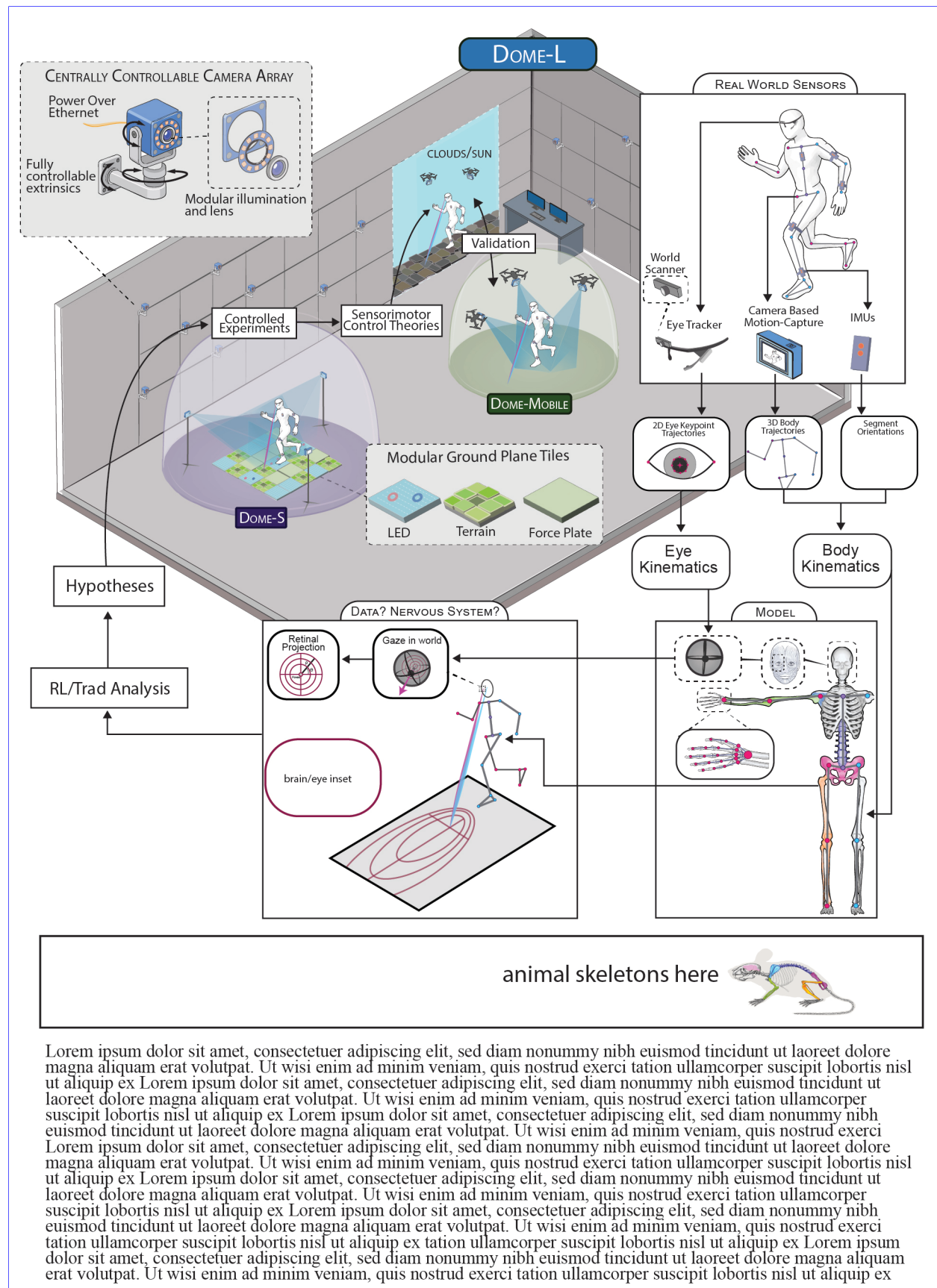
Where durable scientific instruments do exist, they were built by escaping this structure rather than succeeding within it — in institutes engineered for the purpose, or by rare leaders who treated maintenance as first-class against their own career incentives. That escape is the necessary condition. By Conway's Law, an enterprise of domain-named departments, journals, and grants can only produce domain-scoped instruments, reviewed by people who score work inside a single domain — never the long metrology that turns a proof of concept into shared infrastructure. A tool defined by a research domain partitions its users into specialists; a tool defined by a measurement unites everyone who needs that measurement and becomes a commons precisely because it imposes none of their goals. The FreeMoCap Foundation already ran this experiment: scoped to the measurement and refusing to claim a domain, it became a commons of over 15,000 users across 152 countries — but a commons cannot form around abandonware, so **the scoping argument and the engineering argument are the same argument**.

The measurement therefore needs its own organization. The PI left a tenure-track position because the institution could not support this work, and founded the FreeMoCap Foundation as a 501(c)(3) whose primary output is open scientific instruments; the X-Lab's operating model follows directly — scope the deliverable to the instrument, hold full-time career engineers for years so the instrument accumulates the mastery that makes it trustworthy, and fan the science out to a standing network of collaborators who use the tools and feed insight back. This is not a critique offered from the sidelines. We know this failure mode because **we built the thing that does not exhibit it**: a working animal-scale instrument and a global user community,

engineered to be installed, run, and extended by people who did not build it — exactly the standard the X-Labs model exists to fund, and one a university appointment cannot sustain.

1.5. The Vision

[TODO - clean this Vision statement up and fold in some of the cleaner framings from teh bot-text below] The vision of this proposal represents an grand acceleration of movement the FreeMoCap Project has fostered since 2021, and there is no upper limit to our ambition. The global spread of the FreeMoCap software shows that the world has a hunger for this kind of tool, and we intend to feed it. We will put a DOME in every research lab, every classroom, every Physical Therapy clinic, animation studio, and athletic facility. We will create field-deployable wearble DOME-MOBILE systems that provide lab-quality data of humans and animals in the remote wilderness. We will build them into the SCUBA systems of deep sea welders and into the EVA suits and HAB volumes of the ISS. Every country hospital and retirement home will have an AI-enabled, auto-calibrating, self-analyzing Gait and Posture facility that will have plain language conversations with patients about their own kinematic data. We will build DOMES into the sides of moutains to measure the aerodyanmics of hunting hawks, and onto the hulls of shipping boats to understand the mechanics of the dolphins that play in their wake. Every Zoo enclosure will become a 24/7 data stream, with animal tracking models jumpstarted through citizen science and then accelerated through auto-training backprojection pipelines. Previously disconnect specialized results on animal models and niche domains will connect through shared links in a Sensor Grounded Ontology, rather than being scattered across publications in specialized journals. The resulting web of scientific beliefs about neuroscience and biomechanics will trace backwards a shared ontology underpinning a unified science of natural behavior fed by every calibrated DOME, be it a \$1,000,000 flagship facility or \$100 student kit. so that ever reconstructed behavior embodies the same, eternal question - WOULD YOU LIKE TO KNOW MORE?



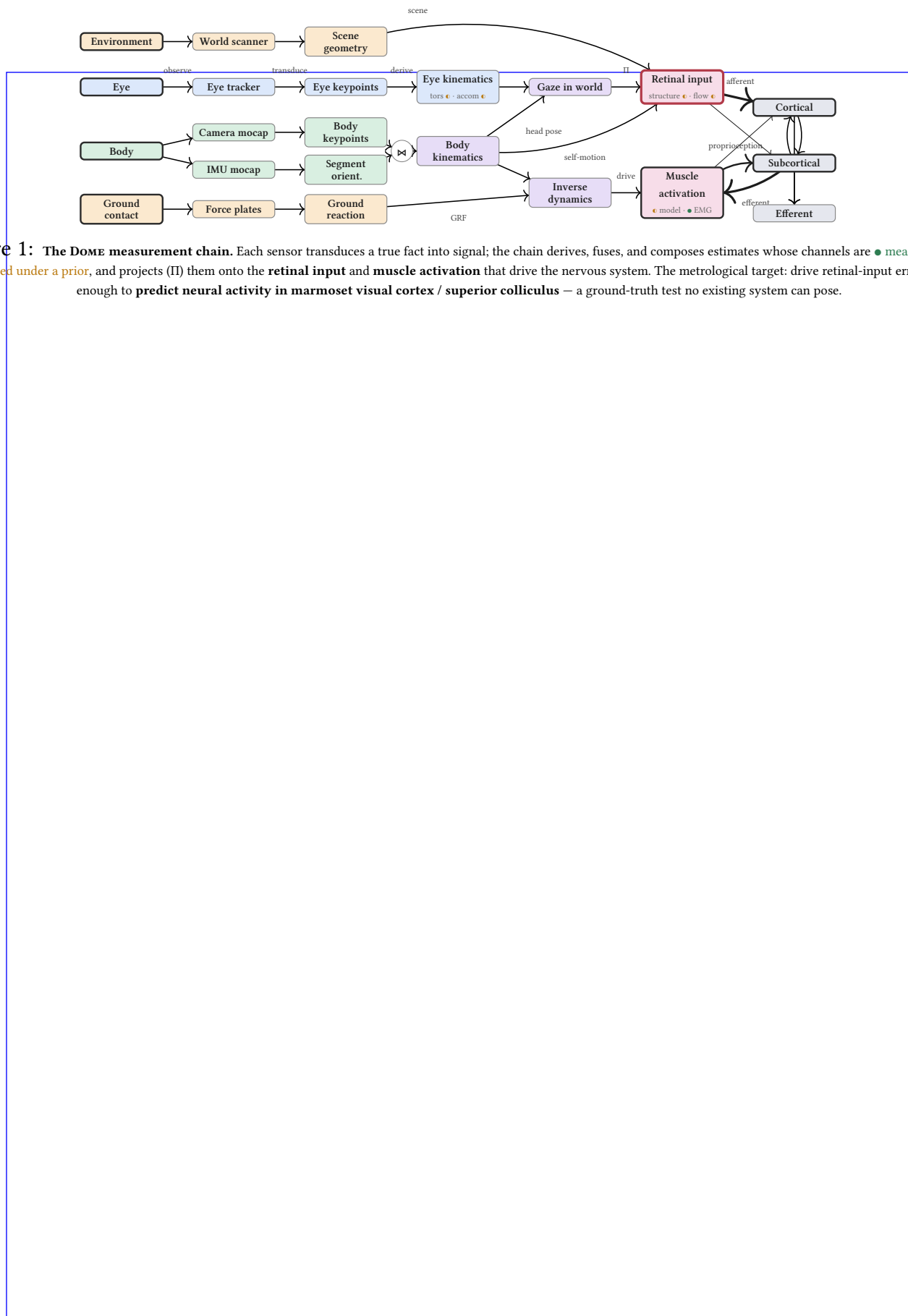


Figure 1: The DOME measurement chain. Each sensor transduces a true fact into signal; the chain derives, fuses, and composes estimates whose channels are ● measured or ● inferred under a prior, and projects (II) them onto the **retinal input** and **muscle activation** that drive the nervous system. The metrological target: drive retinal-input error low enough to **predict neural activity in marmoset visual cortex / superior colliculus** — a ground-truth test no existing system can pose.

Milestone matrix

Item	In place	Phase 0 · 9–12 mo	Phase 1 · 24–36 mo	Phase 2 · variable
ORGANIZATION – OPERATIONAL DELIVERABLES				
Legal & governance	✓ 501(c)(3)	Governance + Autonomy Plan	board seated	
Facility(lease)		lease + move-in	office fit-out	
Full-time team	✓ core FT	hire · onboard	full staff	
Required NSF plans		Data · IP · Security · Gov; Ph1 plan + budget		
IP management	✓ prior IP	IP Mgmt Plan	license · manage	
BODY – TECHNICAL MATURITY				
Camera array		PoC	Prototype	Operational
IMU↔camera fusion	✓ FMC mocap	PoC	Prototype	Validated
EYE – TECHNICAL MATURITY				
Eye tracker (torsion)		Breadboard	Prototype	Operational
WORLD – TECHNICAL MATURITY				
World-scanner		Concept	Prototype	
Floor panels (force/LED)		PoC	Prototype	
ARGPvβ	✓ ARG Pv1		Prototype	Validated
Drone swarm			Concept	PoC
INTEGRATION – FIRST-ORDER				
DOME-L (Auto)		PoC / testbed	Prototype	Operational
DOME-S (Static)	✓ FMC webcam	Breadboard	Validated	Operational
DOME-MOBILE		Concept	Prototype	Validated
DOME-ANIMALS	✓ ferret (6) · mouse (3)	marmoset	guinea fowl	rat

Figure 1: Maturity ladder (TRL): Concept 2 · PoC 3 · Breadboard 4 · Prototype 5–6 · Validated 7 · Operational 8–9 (hardware analogue of PoC→MVP→α→β→release). Org rows = operational deliverables. ✓ = in place pre-award; DOME-ANIMALS cells = per-species level. Each Progress Milestone needs a quantitative exit criterion (FOA §9.2). Stages illustrative.

2. Technology Landscape

2.1. The measurement threads and their instruments

- The DOME unifies six measurement threads, each served by a mature but isolated instrument ecosystem.
 - Kinematics: markerless pose estimation (DeepLabCut, SLEAP, VitPose, FreeMoCap), marker-based (Vicon, Qualisys, OptiTrack), and inertial (Xsens, Notch), with a gap between clinically-valid kinematics and inverse-dynamics-grade accuracy.
 - Kinetics: force plates (AMTI, Bertec), pressure-sensing insoles, and instrumented treadmills, with ground reaction forces still hard to measure outside the lab.
 - Eye tracking: Pupil Labs (Core, Neon, Invisible), Tobii, and SMI, none of which capture torsion or accommodation.
 - Neural recording: Neuropixels, miniscopes (1-photon calcium imaging), and mobile EEG, with neural activity still hard to record during unconstrained movement.
 - Muscle activity: surface and intramuscular EMG at motor-unit resolution, with a gap between what OpenSim models need and what current sensors deliver.
 - Environment: photogrammetry, Gaussian splatting, neural radiance fields, RGB-D (stereo, LiDAR, structured IR), and event cameras.

2.2. The composition problem

- These instruments do not compose, for four concrete reasons.
 - Coordinate frames are incompatible, because each sensor defines its own.
 - Clocks are unsynchronized, because there is no common timebase.
 - Semantic schemes are non-interoperable: a “gaze point” in one system is not the same quantity as in another.
 - Uncertainty does not propagate across modality boundaries.
- Mobile Brain/Body Imaging (Makeig et al. 2009) is the fifteen-year case study: a methodology, not a platform, because the integration burden falls on each lab.

2.3. The metrology gap

- The field co-records without co-measuring.
 - Calibration quality $\kappa = (\kappa_s, \kappa_t)$ is the fundamental bottleneck. [Reference observability calculus.]
 - The GUM/VIM framework applied to multimodal capture shows why an unbroken chain of traceable calibrations matters — and why no existing system provides one.

2.4. The ontology gap

- Existing tools are sensor-specific rather than measurement-specific.
 - They are built around a particular camera model or force-plate vendor, not around the measurement itself; kinematics is kinematics whether it came from cameras or IMUs.
 - The consequence: data from different labs, species, and equipment generations cannot be compared, pooled, or cumulatively improved.
 - The DOME’s sensor-grounded ontology is the solution: define the measurement, not the sensor.

2.5. The CV field’s plateau

- Computer-vision pose estimation has drifted away from the pixels toward ungroundable targets.
 - Targets such as 3D-from-2D, SMPL mesh fitting, and semantic segmentation lack physical ground truth.

- ▶ Trackers still benchmark against aging COCO data scraped from early-2000s Flickr, and fail on out-of-distribution movement — clinical populations, extreme athletics, animal locomotion.
- ▶ The field needs physical ground truth at scale, which a calibrated multi-camera array provides, along with corpora of the movements it currently cannot track.

2.6. The robot training bottleneck

- Reinforcement learning for legged robots is limited by the sim-to-real gap, not by simulation.
 - ▶ Simulation has advanced rapidly (MuJoCo, Isaac Lab, Genesis), but sim-to-real transfer remains the critical barrier.
 - ▶ The missing resource is ground-truth, uncertainty-tagged, multimodal recordings of real agents in real environments.
 - ▶ Existing datasets (Human3.6M, AMASS, GRAB) are single-modality, lab-constrained, or lack uncertainty quantification.

3. Outcomes

3.1. The instrument

- Our five-to-seven-year outcome is a single instrument — the DOME — realized in three human-scale forms and an animal-scale form that records the complete perception-action loop as one calibrated, uncertainty-tagged measurement and yields the same canonical record regardless of which sensors are present.
 - ▶ DOME-L is the warehouse-scale flagship in Greater Boston; it drives each sub-instrument to its limit and is large enough to physically contain the smaller forms, which makes it the metrological reference for the entire network.
 - ▶ DOME-S extends FreeMoCap's webcam capture into the disseminated lab- and classroom-scale form, and DOME-Mobile is the wearable form that carries the measurement onto natural terrain outside the lab.
 - ▶ Because the instrument is defined by its measurement ontology rather than any one sensor, a markerless-camera estimate and an inertial estimate describe the same kinematics and fuse into a single estimate better than either alone.

3.2. The organizing benchmark

- The benchmark that organizes the entire build is veridical retinal input: reconstructing the world projected onto a moving agent's retina to within 1° at the fovea while it fixates a point on the ground at a 45° angle.
 - ▶ This target is defensible rather than aspirational because it composes two independently measured error sources — joint-angle accuracy already characterized in FreeMoCap [cite: dissertation] and the published errors of commercial eye trackers — into one explicit uncertainty budget.
 - ▶ Closing that budget is what forces the hardware advances below, so we state the number once and build toward it.

3.3. Body — trustworthy kinematics

- On the motor half of the loop we deliver kinematics clean enough for inverse dynamics from markerless capture.
 - ▶ The actuated, self-calibrating camera array places daisy-chained Power-over-Ethernet cameras on programmable mounts that stay metrically calibrated as one instrument while being re-aimed and re-focused from a single console, dissolving the bottleneck that leaves large-volume rigs frozen in place because manual re-calibration is too costly; success is post-reconfiguration calibration quality and the time from sub-volume selection to calibrated capture, against a fixed-calibration reprojection baseline.

- ▶ Camera-to-IMU fusion then combines noisy-but-accurate outside-in estimates with precise-but-drifting inside-out estimates under explicit per-joint uncertainty, validated as joint-torque uncertainty against force-plate ground truth and muscle-force uncertainty against EMG-informed estimates.

3.4. Eye — the flagship transducer

- On the sensory half we build the flagship transducer, a binocular eye tracker that measures degrees of freedom commercial trackers omit.
 - ▶ It resolves ocular torsion from iris texture — the degree of freedom that corrupts the curl of retinal optic flow and a standing research target [matthis2022] — targets microsaccade-scale resolution, and pursues lens accommodation via dual-Purkinje imaging as a stretch goal.
 - ▶ The design bet is a dense multimodal sensor array built like a phone camera bank that trades low profile for data quality, which makes it a camera-quality engineering bet rather than a physics one and keeps unit cost on the commodity-imager curve so next-generation eye tracking can saturate the field the way FreeMoCap did for motion capture.
 - ▶ Benchmarks are sub-15-arcminute eye-in-head accuracy, 1° torsion accuracy, and the composed 1° gaze-in-world target above.

3.5. World — the scene the agent acts in

- Reconstructing what the eye sees requires reconstructing the world it acts in, so a head-mounted world-scanner of stereo-RGB, structured-IR, and LiDAR produces per-frame RGB-D and, over longer horizons, a terrain mesh and a SLAM-corrected head trajectory that fixes inertial drift [muller2023].
 - ▶ The same scanner package mounts on drone-swarm nodes that act as a mobile extension of the camera array, grounding DOME-Mobile's inside-out estimates outdoors where a fixed rig cannot reach, benchmarked as gaze and kinematic uncertainty against the DOME-L reference during co-recorded trials.
 - ▶ Modular floor panels — each a force plate, terrain tile, or LED surface — and the ARGP perturbation platform close the world half of the loop by making the ground itself both sensor and controllable stimulus.

3.6. One record across species

- The measurement crosses species, which is the claim that this reshapes a field rather than sharpening a tool: functionally equivalent instruments at collaborator labs yield the same canonical records as the human instrument.
 - ▶ The animal-scale instrument already exists — the team's ferret and mouse builds integrate three-camera gaze, full-body kinematics, and world cameras in one calibrated system, with neural recordings being added now — and the human, mouse, and ferret builds are being brought to functional equivalence, extending to guinea fowl for musculoskeletal biomechanics with EMG [Collab-MD] and marmoset for primate electrophysiology [Collab-AH/JY].
 - ▶ Success is cross-species measurement comparability under shared calibration and data-exchange protocols, so a reconstructed RetinalInput or RigidBody denotes the same thing whether it came from a human, a ferret, or a marmoset.

3.7. What the record unlocks

- Because every channel is calibrated, synchronized, and uncertainty-traced within one sensor-grounded ontology, the record yields capabilities that would each otherwise demand a bespoke project.

- ▶ The instrument generates its own training signal by back-projecting calibrated 3D reconstructions onto each camera's 2D view, and the resulting reprojection error fine-tunes pose estimators, measured as improvement on out-of-distribution movement — clinical gait, acrobatics, animal locomotion — rather than on a saturated benchmark.
- ▶ The record is natively reinforcement-learning-ready because a DOME export maps directly onto MuJoCo and Isaac Lab rigid-body models with uncertainty intact, so DOME-recorded corpora drive sim-to-real policy transfer and inverse-RL recovery of the control policies underlying observed behavior, testable back in DOME-L via ARGV perturbation.

3.8. Dissemination and validation

- The outcome is a countable, validated footprint rather than a released codebase: [N] validated DOME-S instances spanning human biomechanics, robotics, visual neuroscience, and clinical sites, each cross-validated against the DOME-L reference and shipped with a build guide, parts list, and calibration protocol any lab can follow.
 - ▶ We anchor clinical rigor to FDA 510(k) clearance of the FreeMoCap/DOME instrument for rehabilitation and assessment, letting a regulatory bar drive protocol quality.
 - ▶ Phase activities and maturity targets for every line above appear in the Milestone Matrix (Fig. [X]), whose quantitative exit criteria are the benchmarks named here.

4. Senior/Key Personnel

Under the NSF X-Labs model, Phase 0 is the runway for members to transition to full-time before Phase 1. The PI and CSO are already full-time on FreeMoCap; the others commit to full-time by Phase 1, most having already contributed for years — part-time or volunteer, alongside senior industry and academic roles — before any funding existed.

- **Jonathan Matthis, PhD — President / PI (full-time).** Founded and maintains FreeMoCap; two decades measuring the full perception–action loop. Left a tenure-track professorship (Human Movement Neuroscience, Northeastern) because it could not support the integrated tool-building this work requires. NEI NIH K99/R00 recipient (1,400 citations), with a publication record spanning the full loop, from outdoor gaze-and-gait to real-world retinal-input reconstruction.
- **NR — Chief Executive Officer (→ full-time by Phase 1).** Technology-sector engineering leader and startup co-founder, recruited to run the organization so the PI can focus on the technical Mission. Staff Platform Software Engineer at Rapid7; co-founded and scaled MailLift, a venture-backed API startup. Brings the leadership and operations capacity to build FreeMoCap into an autonomous, independent X-Lab.
- **EI — Chief Technology Officer (→ full-time by Phase 1).** Enterprise software architect (15 years, engineer to Principal Architect / CTO) and FreeMoCap's CTO since 2021, essentially since founding. Concurrent CTO at Chorus Innovations; former Principal Architect at Unqork; distributed-systems and data-engineering depth (Python, Node.js, PostgreSQL, Spark; RDF/SPARQL). Ensures enterprise-grade architecture and the calibrated, semantically-unified data backbone the instrument requires.
- **AC, PhD — Chief Scientific Officer (full-time).** [Details to fill from .env] Dissertation validated FreeMoCap against research-grade optical motion capture (Qualisys), demonstrating clinically valid kinematics from commodity hardware. Expertise in clinical tool validation, bench-to-bedside development, neuroprosthetics, and brain imaging; leads DOME-S development and clinical dissemination.
- **JKL, PhD — Chief AI Officer (→ full-time by Phase 1).** Cognitive scientist and applied-AI architect (Applied AI Architect, SOLID Inc.); FreeMoCap contributor since 2024 and lead developer of SkellyBot, its multi-agent AI assistant (NestJS + LangChain, vector-store

retrieval, multimodal inputs). 15+ years handling sensitive human-subjects and health data under FERPA and IRB – anchors data governance and research security, and builds internal AI for development, support, and education.

- **RR – Chief Financial Officer (→ full-time by Phase 1).** Cooperative- and nonprofit-finance specialist and FreeMoCap's Treasurer / volunteer CFO since 2021, its longest-serving financial officer. Founder of Capital Bookkeeping Cooperative; Controller at Honest Weight Food Co-op; Adjunct Professor of Accounting, University at Albany (MS, Accounting). Runs financial operations and grant management; cooperative-governance experience fits the Foundation's open structure and the X-Labs autonomy requirements.
- **KM, PhD – Project Manager, DOME-Mobile (→ full-time by Phase 1).** Computer-vision engineer, data scientist, and longtime PI co-author whose work maps almost one-to-one onto DOME-Mobile. Built the outdoor eye-tracking + photogrammetry + IMU pipeline the DOME generalizes (Muller et al. 2023, 2024, eLife); currently Senior Data Scientist at SynMax (multi-sensor fusion, cloud MLOps). Holds essentially the complete skillset for DOME-Mobile's core instrumentation.
- **MN – Project Manager, DOME-L (→ full-time by Phase 1).** [Details to fill from .env] Clinical biomechanics, medical-systems design, mechatronics, and multi-modal mocap-lab management; holds the skillset for large-scale DOME-L facility design and operation.

5. Team Capabilities Statement

5.1. Complementary expertise

- The leadership team spans the full stack the DOME requires, a combination no single institution's faculty could assemble.
 - Scientific domain expertise: JSM (perception/action neuroscience), AC (clinical biomechanics).
 - Enterprise software architecture: EI.
 - AI systems: JKL.
 - Hardware and mechatronics: KM (mobile/wearable), MN (large-scale facilities).
 - Organizational operations: NR (CEO), RR (CFO/finance).
 - The combination exists because the FreeMoCap Foundation was built from the ground up to produce integrated scientific instruments.

5.2. Governance during Phase 0 and Phase 1

- The FreeMoCap X-Lab (FMC-X) will operate as an independent project within the FreeMoCap Foundation, a 501(c)(3) nonprofit.
 - The Foundation runs other work (e.g., the ongoing FreeMoCap Project), but the missions almost fully overlap.
 - Leadership is nearly identical between FMC-F and FMC-X, with final decision authority resting with the President/PI at the top of the org chart.
 - Research, partnership, and organizational decisions are made in days, not weeks, with no higher management to consult.

5.3. Autonomy

- The FreeMoCap Foundation is already autonomous as of this submission.
 - Full internal control of funding allocation, research direction, partnership agreements, intellectual property, and hiring, with no parent-institution approval required for any operational decision.
 - Satisfies all conditions of the NSF X-Labs Autonomy Factor Test (§6.1.1) at the time of submission.

5.4. Collaborator network

- A standing, active network of research groups already spans every domain the DOME serves.
 - Members include human perception-and-action researchers, robotics and prosthetics groups, visual neuroscientists studying ferret and mouse, musculoskeletal biomechanists studying guinea fowl, and primate electrophysiologists.
 - The network is real and active, not aspirational, with existing collaborations and shared tooling across multiple sites.
 - The PI architects and holds the instrument-plus-network together; the domain arms are executed by named collaborators.

5.5. Community governance and engagement

- The community sets design and roadmap priorities through structured processes that gather input without stalling on consensus.
 - A Request-for-Comments process modeled on Python's PEP system collects broad input on design decisions, work targets, and roadmap priorities.
 - Semi-annual in-person convening: Workshops (train students and out-of-domain experts), Hackathons (onboard developers and connect normally remote workers), Conferences (share DOME-based research), and a Congress (organizational status, plans, and direction).
 - A Community Grants Program and Developer Fund support the open-source network and serve as a testbed for engagement mechanisms (Eurovision-style voting, DAO mechanisms, and others).

5.6. FreeMoCap as completed pilot

- FreeMoCap has already proven this model at smaller scale.
 - It is an open-source markerless motion-capture instrument, built and maintained outside academia, scoped to the measurement — turning cameras and light into calibrated, usable skeleton estimates with an obsessive focus on usability — and refusing to claim any research domain.
 - That refusal is what made it a commons.
 - Receipts: over 15,000 users from 152 countries, over 10,000 GitHub stars, and over 3,500 Discord members.
 - The DOME applies this proven model — measurement-scoped, master-built, open, full-time-maintained — to the far larger target of the complete interaction loop, at a scale only dedicated, autonomous, milestone-based X-Labs funding can reach.

5.7. IP management and dissemination strategy

- The IP strategy keeps the core instrumentation open while actively pursuing adoption and commercialization. [To develop — OT contract requires an IP Management Plan by end of Phase 0.]
 - Open-source core instrumentation with permissive licensing for downstream use.
 - IP structured so platform technologies stay widely accessible while the team pursues adoption, commercialization, and ecosystem growth.

Bibliography